

SHIVAM SINGH

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Notice Period: Serving (available within 30 days)

Software Engineer • Full Stack • AI / RAG • HLS Streaming • Session Analytics

PROFESSIONAL SUMMARY

Software Engineer with 3.5+ years of experience building production-grade Angular applications, Node.js backends, and AI-powered systems. Delivered a real HLS video streaming platform (FFmpeg + CMAF + multi-codec), a RAG document-intelligence system (Gemini AI + pgvector), and a framework-agnostic session-replay monitoring tool — all live and publicly accessible at shivamsingh.website. Strong track record across front-end engineering, REST API design, PostgreSQL, and cloud infrastructure (Vercel, Render, Cloudflare R2).

TECHNICAL SKILLS

Frontend: Angular 12–19, TypeScript, JavaScript (ES6+), RxJS, Angular Signals, Lazy Loading, Standalone Components, HTML5, CSS3, SCSS, Responsive Design

Backend: Node.js, Express.js, REST API Design, JWT Authentication, bcryptjs, Middleware, Error Handling

AI / ML: Gemini API, RAG Architecture, Vector Embeddings (3072-dim), pgvector, Semantic Search, Cosine Similarity, Model Fallback Chain, Grounded Generation

Video: FFmpeg, HLS.js, HLS Streaming, CMAF/fMP4, H.264, HEVC (H.265), AV1, Adaptive Bitrate (ABR), NVIDIA NVENC GPU Encoding

Database: PostgreSQL, Neon DB (Serverless), pgvector, SQL Query Design, Parameterized Queries, JSONB

DevOps: Vercel, Render, Cloudflare R2, Neon DB, GitHub, Git, CI/CD, cron-job.org

Monitoring: Session Replay, MutationObserver, History API, Click Tracking, Event Batching, Framework-Agnostic JS Recorders

Other: pdf-parse, multer, uuid, ngx-translate, Chart.js, forkJoin, switchMap, FormArray, Custom Validators, OAuth (Spotify, Apple Music)

PROFESSIONAL EXPERIENCE

Software Engineer | Songdew Media

Dec 2024 – Present • Gurugram, India

- Architected a **6-role artist registration system** (Label, Individual, Band, etc.) with **200+ dynamic FormArray validators** — role-based UI switching with factory pattern
- Built **8-step wizard component library** for complex onboarding flows, reused across 4 product screens
- Integrated **Spotify OAuth and Apple Music API** for artist profile verification and metadata auto-population
- Used **switchMap chains** for dependent API calls and **forkJoin** for 3 parallel API requests in analytics dashboard — eliminated redundant HTTP calls
- Delivered **Chart.js analytics dashboard** with custom plugins for stream analysis across Spotify, Apple Music, YouTube
- Performed **Angular 12 → 13 → 14 → 15 → 16 migration** — zero regression, adopted standalone components and improved bundle size by 18%
- Implemented **ngx-translate** for Hindi, Portuguese, and 4 regional language support across the platform
- Mentored 2 junior developers through code reviews and Angular best-practice sessions

Software Engineer | Cavisson Systems

Jan 2023 – Nov 2024 • Noida, India

- Built **session replay system** for NetVision APM — captured DOM mutations via MutationObserver, fixed nested iframe recording using recursive DOM traversal
- Developed **element inspection module** with visibility filtering using computed styles — enabled precise click-target identification in session recordings
- Implemented **pagination for large session datasets** — reduced page load from 8s to 1.2s for 10,000+ row tables
- Built performance metric dashboards — LCP, FID, CLS, network timing — consumed from **PostgreSQL via REST APIs** built in Java/Spring Boot

- Designed and maintained **REST API integrations** for real-time data streaming from monitoring agents to Angular dashboard
- Contributed to **NetVision APM platform** — an enterprise product used by 50+ clients for application performance monitoring

PORTFOLIO PROJECTS — SHIVAMSINGH.WEBSITE

FreeFlix — HLS Video Streaming Platform — shivamsingh.website/freeflix

May–Jun 2026

- Built a **Netflix-style HLS streaming platform** from scratch — not an embed. Full **FFmpeg transcoding pipeline** producing CMAF/fMP4 format with 5 quality levels: 360p, 720p, 1080p H.264, 1080p HEVC, 1080p AV1
- Used **NVIDIA NVENC GPU encoding** (h264_nvenc, hevc_nvenc) for 10–20x faster transcoding vs CPU. Chose **CMAF over MPEG-TS** for dual HLS/DASH compatibility and DRM readiness
- Hosted HLS chunks on **Cloudflare R2** (zero egress fees vs AWS S3 \$0.09/GB) — critical for cost-effective video delivery at scale
- Configured **HLS.js with tuned ABR** (abrEwmaFastVoD: 2, abrBandWidthFactor: 0.9) for smooth adaptive bitrate switching. Custom quality selector shows codec badges per track
- **Tech:** Angular 19, HLS.js, Node.js, Express, PostgreSQL, JWT, Cloudflare R2, FFmpeg, Vercel, Render

RAG System — AI Document Intelligence — shivamsingh.website/rag

Jun 2026

- Built a **Retrieval Augmented Generation system** — users upload any PDF and get answers with exact source citations. Prevents hallucination via grounded generation
- PDF processing pipeline: text extraction (pdf-parse) → 600-word chunks with 80-word overlap → **Gemini embeddings (3072-dim)** → stored in **pgvector** (PostgreSQL). Semantic search via cosine distance (`<=>` operator)
- Implemented **5-model fallback chain** (Gemini 3.1 Flash Lite → Gemma 4 31B → Gemma 4 26B → Gemini 2.5 Flash Lite → Gemini 3.5 Flash) — automatic failover on rate limits
- Portfolio Assistant feature uses **systemInstruction separation** to prevent prompt leaking — session-based auth (UUID in localStorage, 24h auto-cleanup via cron)
- **Tech:** Angular 19, Node.js, Gemini API, pgvector, PostgreSQL, pdf-parse, multer, Neon DB

Frontend Monitoring Dashboard — shivamsingh.website/monitoring

Jun 2026

- Built a **framework-agnostic session recorder** (vanilla JS, single script tag) — same architecture as Hotjar/Microsoft Clarity. Works in Angular, React, Vue, or plain HTML without SDK
- Recorder captures: clicks (x/y + element), scroll depth, **SPA navigation via History API monkey-patching**, DOM mutations (MutationObserver), JS errors (window.onerror). Events batch-flushed every 5s with keepalive:true
- Angular dashboard (SaaS light theme): session list with app filters, **event timeline scrubber**, navigation path visualization, click map, event list — all from real monitoring data
- Draws directly on **Cavison Systems APM experience** — implemented similar session replay with nested iframe fix via recursive DOM traversal
- **Tech:** Vanilla JS, Angular 19, Node.js, PostgreSQL, JSONB, Render, Neon DB

EDUCATION

B.Tech — Computer Science & Engineering | **GNIOT, Greater Noida** 2019 – 2023 • Greater Noida, India

ADDITIONAL

- **Live Portfolio:** shivamsingh.website — 3 live projects + AI assistant that answers recruiter questions in real time
- **GitHub:** github.com/shivamaiusage-source — FreeFlix backend, portfolio shell repositories
- **LinkedIn:** linkedin.com/in/shivam-singh-295252342